

Artificial Intelligence in Education: Foundations and Applications

1. Introduction. Artificial Intelligence (AI) is transforming education by enabling personalized learning, adaptive assessment, and intelligent tutoring systems. Modern platforms combine retrieval-augmented generation (RAG) with large language models to answer student questions based on their lecture materials.

2. Core Techniques. The core AI techniques used in educational platforms include natural language processing for summarization, question generation from source material, machine translation of academic texts, and automated proofreading with grammar and spelling correction. These techniques rely on transformer-based models trained on large corpora.

3. Retrieval-Augmented Generation. RAG improves answer accuracy by first retrieving relevant text chunks from student documents, then generating grounded responses with citations. This reduces hallucination and keeps answers faithful to the provided sources. Chunk size and overlap are tuned for balance between context and precision.

4. Quiz Generation. Question generation converts lecture content into multiple-choice questions with plausible distractors. Difficulty calibration depends on the level of reasoning required: recall questions test factual knowledge, while application questions test problem-solving skills.

5. Learning Assessment. Automated assessment provides immediate feedback, tracks student progress over time, and identifies knowledge gaps. Analytics dashboards aggregate usage metrics such as tokens consumed, documents processed, and questions answered.

6. Conclusion. AI-powered educational platforms empower students and researchers to study more effectively. Future improvements include multimodal understanding, voice interfaces, and deeper integration with university learning management systems.

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